

SURIYAN DHANAPAL

Software Engineer | Backend & AI Integration

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<https://github.com/iamsuriyan> | <https://iamsuriyan.github.io/Portfolio/>

SUMMARY

Backend-focused Software Engineer with around 2 years shipping production systems in authentication, real-time infrastructure, and AI/LLM integration. Built and scaled a real-time voice-AI provider layer integrating 25+ STT, TTS, and LLM services behind a unified connector interface, and owned core backend systems — auth, knowledge ingestion, webhooks, and payments. Comfortable taking ambiguous, infrastructure-heavy problems from design to production.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Frameworks: Node.js, Express, Next.js, React, React Native (Expo), FastAPI

Backend & System Design: REST API design, distributed webhook delivery, idempotency & retry design, OAuth 2.0, JWT auth, role-based access control, MongoDB, PostgreSQL, Redis, WebSockets, WebRTC/SIP basics, gRPC

AI & Integration: LLM integration, embeddings, RAG/retrieval, vector search (Weaviate), speech-to-text/text-to-speech pipelines, STS, OCR, browser automation

Tools & Practices: Git, Docker, GCP, AWS (EC2), GitHub Actions (CI/CD), end-to-end feature testing, smoke testing (Claude Code-assisted), Claude Code

PROFESSIONAL EXPERIENCE

Software Engineer — Mindgryd Software Private Limited

Jun 2025 – Aug 2026

AI agent platform · Coimbatore, India

Workflow Automation Platform

Tech: Next.js, TypeScript, Redux Toolkit, Tailwind CSS · FastAPI (Python), OAuth 2.0, Redis, RAG, Weaviate, embedding models, GCP · Node.js/Express, MongoDB, Cassandra, AWS EC2

- Built the Next.js frontend for configuring and managing AI agents and automated workflows, including permission controls for how much autonomy an agent has before an action requires human approval.
- Integrated Razorpay and PayPal, including checkout flows, webhook signature verification, and idempotent transaction handling, and designed the core MongoDB schemas powering the payment layer.
- Built a workspace-specific agent recommendation system: embedded workspace details captured during onboarding and indexed them in Weaviate, then retrieved and ranked relevant agents at search time from a pool of thousands of personalized agents.

AI Memory & Context Platform

Tech: FastAPI (Python), PostgreSQL, pgvector, SQLAlchemy, embeddings, RAG · Next.js frontend

- Built the authentication and onboarding system: OAuth 2.0 sign-in with Google, GitHub, and Microsoft plus email/password, JWT session handling, role-based access control, and automatic organization provisioning on first sign-in.
- Built the knowledge-ingestion subsystem (backend and frontend) with 8 connectors, including Google Drive, Dropbox, Notion, and SharePoint, handling per-user OAuth token storage/refresh and content normalization.
- Built a webhook delivery service with HMAC-signed payloads, at-least-once delivery via idempotency keys, and exponential-backoff retries, eliminating duplicate-processing incidents for downstream integrators.

Real-time Voice-AI Agents Platform

Tech: Vite + React, FastAPI (Python), WebRTC, SIP, gRPC, PostgreSQL, Redis, FlowCat (in-house open-source real-time media engine)

- Integrated 25+ STT, TTS, STS, and LLM providers (OpenAI, Sarvam, Whisper, ElevenLabs, Groq, Gemini, Vertex AI, and others) behind a unified connector interface on FlowCat, our in-house, open-sourced real-time media engine, cutting new-provider integration time from days to hours.
- Debugged provider-specific failures affecting live calls (gRPC streaming, tool-call mapping, API-key resolution) and built provider-aware test flows to catch regressions before release.
- Built a Vite + React dashboard for real-time monitoring and debugging of live voice-AI calls, replacing raw-log digging with a single view of call status, provider health, and test-flow results — the team's go-to tool for triaging live call issues.

Software Engineer — Hifresh Agro Private Limited

Sept 2024 – Jun 2025

Agro-tech product company · Coimbatore, India (same founding team; moved to Mindgryd when it was spun up)

Tech: React Native (Expo), Redux · Node.js/Express · Zoho Books API, ESC/POS thermal printers, LRU caching

- Built and owned an internal delivery-management mobile app (React Native / Expo) used daily by the company's delivery team end to end, extending the existing Node.js/Express backend.
- Developed APIs for delivery assignment, billing, and invoicing (Zoho Books integration) and integrated ESC/POS thermal printers for in-app receipt printing.
- Added response caching and structured logging to the Express backend's highest-traffic endpoints, eliminating redundant database queries on the app's most-used routes.

CONTRACT / PROJECT WORK

Cooper Compass — Document & Email Intelligence for Logistics (IIT Bombay-incubated startup)

coopercompass.com · Node.js, Express, TypeScript, OpenAI, Google Vision OCR, Tesseract.js, OAuth 2.0 (Gmail / Microsoft Graph), PostgreSQL, Redis, Docker, GitHub Actions

- Built an email-intelligence platform that cut manual extraction of job/BL numbers from shipping emails from ~2 hours to ~8 minutes per shipment across 20+ pilot users.
- Built real-time email ingestion from Gmail and Outlook (OAuth 2.0 with Gmail Pub/Sub and Microsoft Graph push notifications), with Server-Sent Events driving live dashboard updates.
- Implemented the document-extraction pipeline: Google Vision OCR with a Tesseract.js fallback, OpenAI-based classification, and a regex validation layer for job, BL, AWB, and port fields, auto-grouping documents by shipment.

EDUCATION

Bachelor of Engineering — Electronics and Communication Engineering

2019 – 2023

Dr. N.G.P. Institute of Technology